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DaNet: Domain-adaptive white blood cell classification through synthetic augmentation and cross-domain feature alignment

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ABSTRACT

Background and Objective: Automated classification of white blood cells (WBCs) plays a vital role in improving clinical diagnostics and disease monitoring. However, current methods frequently face challenges with generalization, as they depend on training and testing data drawn from the same distribution. This limitation hinders their effectiveness in real-world clinical settings.

Methods: We introduce DaNet, an innovative domain-adaptive method for classifying WBCs that leverages domain generalization techniques. DaNet comprises two key components: Balanced Multi-Domain Mixup (BMDM) and Data Distribution Alignment (DDA). BMDM serves as a data augmentation technique specifically designed to address the high similarity and class imbalance inherent in WBC datasets. By generating synthetic data that captures more distinctive and discriminative features, BMDM enhances the model's ability to learn robust representations. DDA further aligns these features across different domains, enabling the model to learn domain-invariant characteristics.

Novelty: The proposed BMDM facilitates a more continuous latent space for domain-invariant features across multiple source domains, which effectively alleviates the alignment challenges commonly encountered by DDA-based methods in WBC image classification tasks, particularly in the presence of subtle inter-class differences and class imbalance.

Results: Extensive experiments demonstrate that DaNet performs reliably in WBCs classification tasks, particularly excelling in cross-domain generalization. The method shows solid performance across various blood cell classification tasks, indicating its effectiveness. Its innovative approach to data augmentation and domain alignment enhances the model's robustness and generalizability.

1. Introduction

White blood cells (WBCs), also known as leukocytes, are a critical component of the human immune system. They play a vital role in defending the body against invading pathogens, such as bacteria, viruses, parasites, and fungi [1]. WBCs can be primarily classified into five main types of neutrophils, eosinophils, basophils, monocytes, and lymphocytes, each type being with distinct functions in the body's immune response. Changes in the number or proportions of these different WBC types can often indicate the presence of certain diseases or health issues [2]. For example, an increase in lymphocytes may suggest that a patient has leukemia, a hypersensitivity reaction, or a viral infection [3,4]. Therefore, accurately identifying and counting different types of WBCs can provide crucial insights for diagnosing blood-related diseases. Currently, the analysis of these diseases heavily relies on skilled professionals manually inspecting blood smears under a microscope. This process is inherently tedious and time-consuming, with the potential for errors in the detection of WBCs. Such errors can significantly impact the optimal treatment regimen for patients. The emergence and rapid development of deep learning (DL) have revolutionized the field of medical image analysis, particularly in the realm of automated classification of blood cell images. DL-based techniques have effectively supported disease inspection and diagnosis in various medical imaging tasks, demonstrating their potential to streamline and enhance the accuracy of WBC identification, and consequently, facilitating the diagnosis of blood-related disorders [5,6].

Currently, research on WBC classification tasks in DL primarily focuses on convolutional neural networks (CNNs) [7] and vision transformers (ViTs) [8]. While ViTs excel at capturing global information [9,10], the advantage of CNNs lies in their low computational and spatial complexity, making them widely used in WBC classification tasks [11–14]. In CNN-based approaches, the WBC classification task can be

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broadly categorized into two research directions: in-domain generalization and out-of-domain generalization. The main challenges of in-domain generalization for WBC classification tasks stem from issues inherent in WBC data. For example, the subtle differences between WBC classes result in low inter-class distinguishability [15,16], and there is a prevalent issue of data imbalance in WBC datasets [17]. Data imbalance refers to significant differences in the number of samples across different classes, leading to learning outcomes that are biased toward predicting the majority classes while performing poorly on minority classes.

Most current WBC research focuses on in-domain generalization, where significant breakthroughs have already been achieved in WBC classification tasks. In contrast to in-domain generalization, out-of-domain generalization does not rely on the independent and identically distributed (IID) assumption, which posits that training and testing data follow the same distribution. In real-world scenarios, trained models often need to generalize to unseen distributions. For instance, a blood cell classification model trained on previously acquired images may achieve high accuracy under a consistent data distribution. However, changes in the hospital's sampling procedures — such as modifications in staining protocols or microscope settings — can lead to a significant decline in model performance [18]. Consequently, models based on the IID assumption may perform well on data from the same distribution but poorly on unseen data distributions. To better meet practical needs for automated WBC classification, domain generalization (DG) techniques have received significant attention in recent years [19]. The idea of DG is to learn from one or multiple training domains to develop a domain-agnostic model that can be applied to unseen domains [20]. For instance, Li et al. designed a local alignment constraint to encourage the model to capture domain-invariant features [21]. Chang et al. [22] improved DG by introducing stain mix-up for data augmentation, thereby reducing domain gaps caused by color variations between different sources. Additionally, Umer et al. [23] employed the Balanced Domain-Class Distribution Alignment (BoDA) [24] algorithm, which is used for multi-source domain feature alignment and is particularly effective in handling imbalanced data.

In this paper, we focus on enhancing the domain generalization capability of WBC classification tasks and propose DaNet method to improve DG performance. The DaNet architecture comprises two key components: Balanced Multi-Domain Mixup (BMDM) and Data Distribution Alignment (DDA). Due to the data imbalance in WBC datasets and the subtle differences between classes, general CNN methods struggle to extract distinguishing features for each class. Inspired by Balanced-Mixup [25], we introduced the concept of Balanced-Mixup into multiple source domains, forming BMDM for data augmentation. BMDM effectively augments and balances samples during model training, enabling the model to learn more distinctive features. DDA is a method for improving DG in classification tasks, used to align the features extracted from multiple source domains. We have enhanced the BoDA algorithm to formulate an innovative loss function tailored for mixed features, thereby grounding our implementation of the DDA method. The loss function not only bolsters the intra-class similarity among features originating from diverse source domains but also accentuates the inter-class differences, enabling the establishment of clearer decision boundaries [26]. We conducted experiments on three publicly available WBC datasets. In comparison to recent state-of-the-art cross-domain techniques, our proposed DaNet demonstrates superior overall performance in generalizing WBC classification across domains, highlighting its effectiveness in addressing cross-domain challenges. The primary contributions of this paper are in three-fold:

- We propose a novel domain-aligned network named DaNet to improve the cross-domain generalization ability in WBC classification tasks, which better aligns features of different classes across multiple source domains, establishing clearer decision boundaries.
- We propose a balanced multi-domain mixup scheme, *i.e.*, BMDM, which ensures that domain-invariant features from different source domains occupy a more continuous latent space, facilitating the learning of domain-specific distinguishing features.
- We conduct extensive experiments on three publicly available WBC datasets, robustly validating that DaNet exhibits superior cross-domain generalization and adaptability compared to state-of-the-art domain generalization methods.

The remainder of this paper is organized as follows. Section 2 provides a comprehensive review of related research on out-of-domain generalization in medical image analysis. Section 3 specifically introduces the proposed DaNet method, providing a detailed overview of the key components and architecture of the model. The experiment results are then discussed in-depth in Section 4, including an analysis of the performance on various benchmark datasets and tasks. Finally, the paper concludes in Section 5 with a summary of the key findings and potential directions for future research.

2. Related work

In this section, we review studies related to out-of-domain generalization in medical image analysis. These works are broadly categorized into two typical approaches: Domain Adaptation (DA) and Domain Generalization (DG). In addition, Table 1 provides a clearer summary of several representative out-of-domain methods, including their categorization as well as a comparison of their advantages and limitations.

2.1. Domain adaptation

Domain Adaptation (DA) in medical image analysis aims to adapt models trained on a source domain to perform effectively on a target domain that differs in imaging modalities, acquisition protocols, or patient populations. The primary objective of DA is to mitigate the domain shift between the source and target domains [27]. To address this challenge, various DA strategies have been proposed. A significant portion of existing research focuses on **domain alignment**, which seeks to align global or class-wise feature distributions between the source and target domains. Representative approaches include methods based on maximum mean discrepancy (MMD) [28] and regularization-based [29] alignment strategies, such as GroupDRO [30]. Another prominent line of research involves **adversarial learning**, which introduces a domain discriminator and utilizes adversarial training to encourage the learning of domain-invariant feature representations. Many of these techniques build upon the Generative Adversarial Network (GAN) [31] framework. For instance, DM-ADA [32] enhances the traditional GAN approach by introducing Domain-Mixup strategies at both pixel and feature levels. Similarly, Co-UNet-GAN [33] and U-shaped-GAN [34] incorporate U-Net architectures to further capture multi-scale features.

2.2. Domain generalization

In medical image analysis, domain generalization (DG) aims to train models on multiple heterogeneous source domains such that they generalize well to previously unseen target domains. Unlike domain adaptation (DA), DG does not utilize any target domain data during training, making it more challenging due to the absence of explicit supervision to address domain shifts. Existing DG techniques can be broadly categorized into three main groups: meta-learning, domain alignment, and data augmentation. **Meta-learning** methods [35–37] simulate domain shifts by creating meta-training and meta-testing splits, enabling the model to learn how to generalize across domains. A representative approach is MLDG [38], which synthesizes virtual testing domains within each mini-batch to emulate training/testing

Table 1
Summary of the advantages and limitations of several representative out-of-domain generalization method.

Method	Category	Advantages	Limitations
MMD [28]	Domain alignment	Simple and effective; minimizes distributional discrepancy between source domains	Requires appropriate kernel selection; weak in modeling complex structural information.
GAN [31]	Adversarial-based	Capable of modeling complex distributions; can synthesize virtual samples from diverse domains for generalization.	Unstable training; may generate semantically irrelevant samples.
DM-ADA [32]	Adversarial-based	Enhances feature robustness by mixing source domain features in latent space	Requires careful tuning of mixing ratio; may produce unnatural feature combinations.
Group-DRO [30]	Domain alignment	Minimizes worst-case group risk; improves robustness on hard subdomains.	Group/domain definition often relies on prior knowledge; may suffer from unstable optimization.
MLDG [38]	Meta-learning	Simulates domain shifts by meta-training/testing splits; significantly enhances generalization capability.	Computationally expensive; sensitive to task/meta-task formulation.
BoDA [24]	Domain alignment	Emphasizes intra-class feature consistency and inter-class separability; Effectively distinguishes semantic classes in multi-source settings.	Requires accurate class and label annotations and careful tuning of the loss weighting.
LLDG [43]	Domain alignment	Extracts shared features using a linear dependency regularizer.	May be less effective in scenarios with non-linear feature sharing.
MixStyle [44]	Data augmentation	Efficient and unsupervised; mixes instance-level style statistics during training to implicitly align domains.	May not improve discriminative features; less effective for domains with subtle style variations.

domain shifts during model optimization. **Domain alignment** methods [39–42] aim to learn domain-invariant representations by minimizing distributional discrepancies or extracting causal features. For example, BoDA [24] promotes similarity among features from the same class across different domains while penalizing similarity between features from different classes both within and across domains. Similarly, LDDG [43] employs a variational encoding framework combined with a linear dependency regularizer to extract shared features from multi-domain medical images. **Data augmentation** strategies [44–46] improve generalization by increasing the diversity of source domains. For instance, MixStyle [44], proposed by Zhou et al., mixes instance-level feature statistics to simulate domain shifts. In addition, Zhang et al. [47] proposed a deep stacked transformation method for medical image classification, which enhances robustness to domain variations.

3. Methodology

In this section, we provide a detailed discussion of the design of DaNet, as depicted in Fig. 1. We clarify relevant definitions and detail the implementations of the key components, BMDM and DDA, to ensure a clear understanding of DaNet's implementation.

3.1. Definitions and notations

Given a multi-domain classification task of WBC with D source domains, $\mathcal{X}$ be the input space and $\mathcal{C} = \{1, \dots, C\}$ be the discrete label space. Let $\mathcal{S}^j = \{(\boldsymbol{x}_i^j, c_i^j)\}_{i=1}^{N_j}$ be the jth domain training set($1 \le j \le D$), where $\boldsymbol{x}_i^j \in \mathbb{R}^n$ is an input, c_i^j is the label of $\boldsymbol{x}_i^j$, and N_j is the number of samples in $\mathcal{S}^j$. The union of all the training sets is defined to be $\mathcal{S}^* = \bigcup_{j=1}^{D} \mathcal{S}^j$. Let $\mathcal{S}^{j,c}$ be the set of samples from $\mathcal{S}^j$ with class label being c, that is $\mathcal{S}^{j,c} = \{(\boldsymbol{x}^j, c^j) \in \mathcal{S}^j : c^j = c\}$ and let $N_{j,c} = |\mathcal{S}^{j,c}|$ be the number of samples in $\mathcal{S}^{j,c}$. For convenience, we use $\mathcal{M} = \mathcal{D} \times \mathcal{C} := \{(j, c) : d \in \mathcal{D}, c \in \mathcal{C}\}$ to denote the set of all domain-class pairs that occurs in $\mathcal{S}^*$, where $\mathcal{D} = \{1, 2, \dots, D\}$. Whenever a sample $\boldsymbol{x}$ with domain-class pair (j, c) is mentioned, it is always means that $\boldsymbol{x}$ has a class label c and originates from the jth domain.

Denote by $\mathcal{Z} \subseteq \mathbb{R}^m$ the representation space and let $f : \mathcal{X} \to \mathcal{Z}$ be a mapping with feature $\boldsymbol{z} = f(\boldsymbol{x})$ being the representation of input $\boldsymbol{x}$. The mean of feature representations corresponding to samples with domain-class pair (j, c) is denoted by $\boldsymbol{\mu}^{j,c}$ and the overall mean of feature representations across all domains is denoted by $\boldsymbol{\mu}$. The final prediction $\bar{c} = g(\boldsymbol{z})$ is determined by a classification function $g : \mathcal{Z} \to \mathcal{C}$. Additionally, to compute the dissimilarity between two feature vectors, we introduce the Euclidean distance $\tilde{\mathrm{d}}(\boldsymbol{x}, \boldsymbol{y}) = \sqrt{(\boldsymbol{x} - \boldsymbol{y})^{\mathrm{T}}(\boldsymbol{x} - \boldsymbol{y})}$, where $\boldsymbol{x}$ and $\boldsymbol{y}$ are two vectors of the same length.

3.2. Overview

In many studies, it has been strongly demonstrated that aligning the data distributions of different source domains can help improve the generalization capability of the model [48,49]. However, the subtle differences among the WBC classes and the issue of data imbalance make it challenging for DDA methods to perform optimally. To address this, we propose DaNet, which aims to establish clearer decision boundaries for WBC classification. DaNet consists of two key components: BMDM and DDA, as illustrated in Fig. 1. BMDM is a data augmentation technique. We first employ BMDM to create a more balanced and diverse sample distribution. This enables the Featurizer (e.g., ResNet50) to learn more discriminative features across the different source domains. DDA further aligns the intra-class and inter-class relationships of the features, allowing the network to eliminate redundant parameters and extract domain-invariant features across multiple domains. Therefore, we then apply DDA intervention to the features from the Featurizer and incorporate it into the loss function. Finally, by combining the aligned loss and the predicted loss, the network weights are updated through Backward Propagation.

To thoroughly explain the implementation details of DaNet, we will next introduce some related notations, followed by a detailed description of the implementation of BMDM and DDA.

3.3. Balanced multi-domain mixup

Inspired by Balanced-Mixup [25], which combines Mixup Galdran et al. and sampling strategies to achieve strong generalization performance in imbalanced medical image classification tasks, we propose a data augmentation component called BMDM, applicable to multi-source domains. As shown in Fig. 1, BMDM firstly employs instance-based sampling on each domain $\mathcal{S}^j$ to obtain a batch $\mathcal{B}^j$, and employs class-based sampling on the union $\mathcal{S}^*$ of all domains to get another one batch, denoted by $\mathcal{B}_j^*$. Note that all the batches $\mathcal{B}^j$ and $\mathcal{B}_j^*$ have the same size. Then the batches $\mathcal{B}^j$ and $\mathcal{B}_j^*$ are mixed by using the Mixup technique to form a new batch $\hat{\mathcal{B}}^j$. Specifically, in accordance with the order of sampling, a sample is taken from each of the two batches, and they are then paired and mixed according to the following formula:

$$\hat{\boldsymbol{x}}^j = (1 - \gamma)\boldsymbol{x}^j + \gamma \boldsymbol{x}^*, \quad \hat{c}^j = (1 - \gamma)c^j + \gamma c^* \tag{1}$$

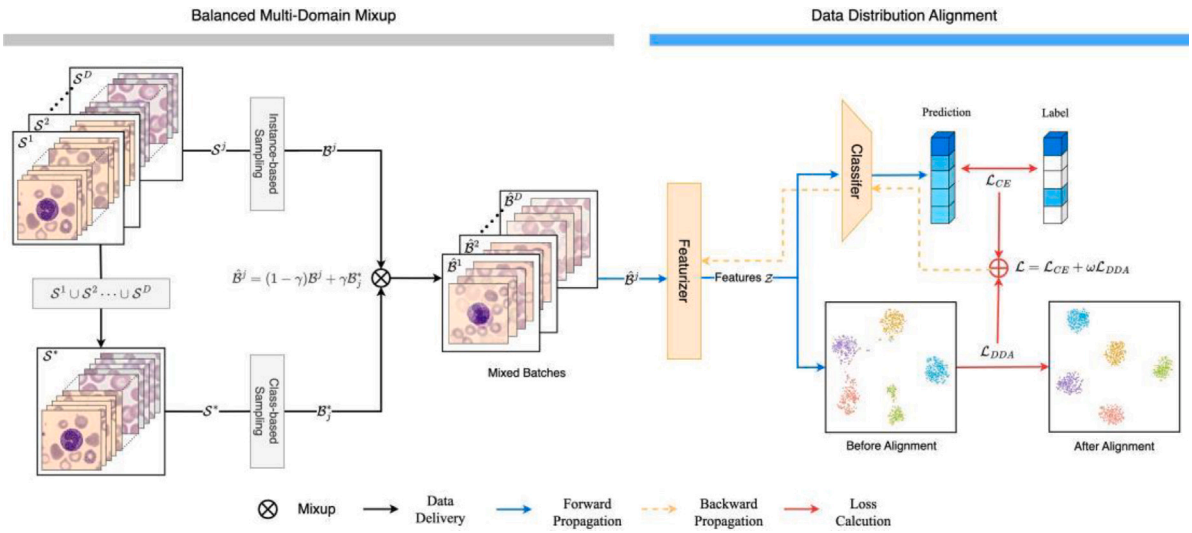

Fig. 1. The implementation of the DaNet involving two main components. The first component is the Balanced Multi-Domain Mixup (BMDM), which balances and mixes the dataset, allowing the Featurizer (e.g., ResNet50) to better extract distinctive features from WBC images. The second component involves calculating the DaNet loss ($\mathcal{L}_{DDA}$) based on the extracted features, reducing the discrepancies among the same-class samples and increasing the distinctions between different-class samples. The classifier used in our experiments is a fully connected (FC) layer following a Softmax layer.

where $(\boldsymbol{x}^j, c^j) \in \mathcal{B}^j$, $(\boldsymbol{x}^*, c^*) \in \mathcal{B}^*_j$, $(\hat{\boldsymbol{x}}^j, \hat{c}^j) \in \hat{\mathcal{B}}^j$ and $\gamma \sim Beta(\alpha, 1)$. Here $Beta(\alpha, 1)$ is an exponential-like distribution, which deliberately accentuating the prevalence of samples derived from $\mathcal{S}^d$. Regarding sampling strategies, instance-based sampling selects data instances in proportion to their occurrence in the dataset, using random selection without class stratification. Conversely, class-based sampling prioritizes equalizing selection probabilities across classes. In short, BMDM optimizes the balance across different source domains and facilitates cross-domain Mixup to generate diverse samples, enabling the model to more easily learn distinctive features of different classes.

3.4. Data distribution alignment

In this section, we describe the details of the data distribution alignment technique used in DaNet. As shown in Eq. (1), the sample $\hat{\boldsymbol{x}}^j$ from $\hat{\mathcal{B}}^j$ is synthesized by samples $\boldsymbol{x}^j$ with domain-class pair (j, c^j) and $\boldsymbol{x}^*$ with domain-class pair (j^*, c^*). For the sake of simplicity in notation and without causing confusion, we will denote c^j as c. The Featurizer (e.g., ResNet-50) extracts features from the synthesized sample $\hat{\boldsymbol{x}}^j$, resulting in the feature representation $\boldsymbol{z} = f(\hat{\boldsymbol{x}}^j)$. The loss function on $\boldsymbol{x}^j$ under distribution alignment operation is defined to be

$$\tilde{\mathcal{L}}_{BoDA}(\boldsymbol{z}, j, c, \{\boldsymbol{\mu}\}) = \frac{-1}{|\mathcal{D}|-1} \sum_{k \in \mathcal{D}\setminus\{j\}} \log \frac{\exp\left(-\lambda^{k,c}_{j,c} \tilde{\mathrm{d}}\left(\boldsymbol{z}, \boldsymbol{\mu}^{k,c}\right)\right)}{\sum_{(j',c') \in \mathcal{M}\setminus(j,c)} \exp\left(-\lambda^{j',c'}_{j,c} \tilde{\mathrm{d}}\left(\boldsymbol{z}, \boldsymbol{\mu}^{j',c'}\right)\right)} \tag{2}$$

where $\lambda^{j',c'}_{j,c} = \left(\frac{N_{j',c'}}{N_{j,c}}\right)^{\nu}$ is a calibration parameter used for balancing and ν is a hyperparameter used to control the influence of the calibration parameter. In Eq. (2), the numerator component represents the differences within the same-class samples, which ought to be minimized, whereas the denominator component represents the differences between different-class samples, which need to be amplified. From an overall perspective, the design of the BoDA algorithm is similar to common clustering algorithms [50], with $\boldsymbol{\mu}$ acting as the centroid, pulling features of the same class closer together and pushing features of different classes further apart. This enables the model to establish clear decision boundaries for WBC classes.

In our DaNet method design, the DDA operation is applied to the feature space of the augmented samples generated by BMDM, as shown in Fig. 1. Since BMDM's augmented samples may belong to different domain-class pairs, the alignment must account for these differences to ensure consistency between the samples and their features. The loss function for DDA is designed to be

$$\mathcal{L}_{DDA} = (1-\gamma)\tilde{\mathcal{L}}_{BoDA}(\boldsymbol{z}, j, c, \{\boldsymbol{\mu}\}) + \gamma \tilde{\mathcal{L}}_{BoDA}(\boldsymbol{z}, j^*, c^*, \{\boldsymbol{\mu}\}) \tag{3}$$

where (j, c) and (j^*, c^*) are the domain-class pairs of $\boldsymbol{x}$ and $\boldsymbol{x}^*$, respectively.

In order to measure the differences between the model's prediction labels and the true labels, we introduce the concept of Cross-Entropy (CE) loss function. The standard CE loss function is defined as:

$$\mathcal{L}_{CE}(\hat{c}, \bar{c}) = -\sum_{i=1}^{D} \hat{c}_i \log\left(\bar{c}_i\right), \tag{4}$$

where all labels are encoded using one-hot encoding and $\hat{c} = (\hat{c}_1, \hat{c}_2 \ldots, \hat{c}_D)$ and $\bar{c} = (\bar{c}_1, \bar{c}_2, \ldots, \bar{c}_D)$ represent the true label and predicted label for the corresponding inputs, respectively.

The overall loss function for DaNet is formulated as the weighted combination of $\mathcal{L}_{DDA}$ and $\mathcal{L}_{CE}$, more precisely,

$$\mathcal{L} = \mathcal{L}_{CE} + \omega \mathcal{L}_{DDA}, \tag{5}$$

where ω is a trade-off hyperparameter used to control the DaNet loss. By default, our DaNet method applies instance-based sampling to dataset $\mathcal{S}^m$ and class-based sampling to $\mathcal{S}^*$. To further enhance data balance among the different WBC classes, we swap the sampling strategies of $\mathcal{S}^m$ and $\mathcal{S}^*$, applying class-based sampling and instance-based sampling respectively, and define this method as DaNet*. The purpose of DaNet* is to further improve class balance, making DDA more effective in establishing clearer boundaries for each class.

4. Experiment

In this section, we first describe the implementation details of our experimental design, including the datasets, baselines and some related experimental setups. Then we report the empirical outcomes of ablation study and comparative experiments. Our subsequent analysis endeavors to provide an insightful interpretation of the experimental results, critically examining the implications and nuances that emerge from the data.

4.1. Datasets

Within our experimental framework, we leverage three publicly accessible and widely recognized WBC datasets, namely Matek-19 [51],

Table 2
The description of datasets.

Dataset	Quantity	File format	Dimension (pixels)	Description
Matek-19	10,248	TIFF	400 × 400	Images were taken from peripheral blood smears of 100 patients diagnosed with Acute Myeloid Leukemia at Munich University Hospital.
Acevedo-20	14,822	JPG	360 × 363	Images were acquired using the analyzer CellaVision DM96 in the Core Laboratory at the Hospital Clinic of Barcelona.
Bodzas-23	10,426	JPG	256 × 256	The images capture various lighting conditions and blood film artifacts that can arise from manual blood smear preparation and staining.

Table 3
Quantities of each classes.

	Matek-19	Acevedo-20	Bodzas-23
Basophile	79	1218	1023
Neutrophil	8593	3279	3300
Eosinophil	424	3117	1017
Lymphocyte	3937	1214	3046
Monocyte	1789	1420	2040

Acevedo-20 [52] and Bodzas-23 [53]. The basic information of these datasets is presented in Table 2.

The WBCs in these datasets can be divided into five class labels, namely Neutrophils, Basophils, Monocytes, Eosinophils and Lymphocytes. The quantitative distribution of samples pertaining to each class within each dataset is meticulously outlined in Table 3.

These three datasets have different domain feature distributions, and the Matek-19 dataset exhibits an obvious data imbalance problem. To verify the cross-domain generalization capability of the model, we adopt the leave-one-domain-out method: we only use two of the datasets as source domains, and the remaining one as the unseen test set. For the source domains, we randomly select 80% of the data as the training sets and the remaining 20% as the validation sets. Additionally, our method is based on multi-source domain training models. For single-source domain baselines, we will merge multiple training sets into a single training set for model training (e.g., WBC-Benchmark [18]).

4.2. Baselines

To rigorously evaluate the cross-domain generalization capability of our approach on WBCs, we have curated a selection of the most recent and relevant domain generalization methods from the literature, as outlined below.

- WBC-Benchmark [18]: WBC-Benchmark is a single-source domain method that optimizes batch normalization to adapt the model to the domain shift in the WBC classification task.
- DDG-Mixstyle [36,44]: DDG-Mixstyle is a meta-learning-like DG method that uses a meta-adjuster to twist the network parameters, and employs Mixstyle as a data augmentation technique.
- BoDA [24]: BoDA is a distribution alignment-based DG method that tracks the upper bound of transferability statistics and ensures balanced alignment across imbalanced domain-class distributions.
- DM [32]: DM is a domain adaptation method based on domain mixup, which is jointly conducted on pixel and feature level to improve the robustness of models across multiple domains.
- MTL [45]: MTL is a transfer learning-based DG method that augments the original feature space with the marginal distribution of feature vectors.
- IRM [40]: IRM learns domain-invariant representations by constraining the invariant predictive risk.

Table 4
Results on Matek-19 dataset.

Method	F1-micro (%)	F1-macro (%)
WBC-Benchmark [18]	95.33 ± 0.86	80.81 ± 1.97
DDG-Mixstyle [36]	**96.33 ± 0.21**	**84.30 ± 1.68**
BoDA [24]	95.62 ± 0.79	82.77 ± 1.31
DM [32]	95.26 ± 0.76	81.30 ± 1.32
MTL [45]	95.65 ± 0.66	82.42 ± 2.31
IRM [40]	92.38 ± 1.02	73.17 ± 2.76
DaNet	95.80 ± 0.59	82.63 ± 2.03
DaNet*	96.01 ± 0.50	83.01 ± 1.45

The best results are **bold**.

4.3. Experimental setup

All the experiments are conducted on an RTX 3090 GPU with PyTorch 2.1.1. We use ResNet-50 [54] pre-trained on ImageNetV1 as the backbone and the Adam optimizer, with a weight decay of 1e−4. The initial learning rate is set to the default of 5e−5, except when Acevedo-20 and Bodzas-23 are used as the training datasets, in which case it is set to 6e−6. The network optimization is performed with a batch size of 64, 7000 training steps, and input size (C, H, W) of (3, 224, 224).

For the relevant hyperparameters of DaNet, we set the α parameter in $\gamma \sim Beta(\alpha, 1)$ to 0.1. The ω and ν parameters in the loss function (Eq. (5)) are set to 0.1 and 0.5 respectively. We initiate the BoDA algorithm at the 500th step to ensure an appropriate initialization point and use a linear classifier as the decision-making component for the classification task. Additionally, we establish a periodic update mechanism for the mean ($\boldsymbol{\mu}$) of all features, refreshing this critical parameter every 120 steps to enhance the model's adaptability and responsiveness to changes in the data.

In order to ensure the robustness and reliability of the evaluation, all baseline models will undergo a rigorous testing procedure, wherein they are executed ten times independently. Additionally, both the Fl-micro and F1-macro metrics will be assessed on an unseen test set, separately. The results will then be aggregated by calculating the mean and standard deviation (mean ± std) of these performance metrics, providing a comprehensive and statistically sound basis for comparing the various baselines.

4.4. Experiment results

To validate the DG performance of DaNet and DaNet* in WBC classification tasks, we conducted comparative experiments using the specified datasets and experimental setup. The baseline comparison results, with Matek-19, Acevedo-20, and Bodzas-23 as unseen test sets, are presented in Tables 4, 5, and 6, respectively. Table 7 provides an average summary of these results. From these tables, we observe the following:

1. Both DaNet and DaNet* outperform prior state-of-the-art DG methods, achieving the best results in terms of average F1-micro

Table 5
Results on Acevedo-20 dataset.

Method	F1-micro (%)	F1-macro (%)
WBC-Benchmark [18]	83.88 ± 4.60	80.33 ± 6.53
DDG-Mixstyle [36]	84.00 ± 4.15	82.67 ± 4.83
BoDA [24]	92.37 ± 0.97	90.29 ± 0.92
DM [32]	89.59 ± 2.81	86.92 ± 3.24
MTL [45]	90.82 ± 4.24	89.25 ± 3.98
IRM [40]	67.89 ± 10.85	59.19 ± 10.41
DaNet	**93.47 ± 1.79**	**92.17 ± 2.20**
DaNet*	93.26 ± 1.01	91.58 ± 1.23

The best results are **bold**.

Table 6
Results on Bodzas-23 dataset.

Method	F1-micro (%)	F1-macro (%)
WBC-Benchmark [18]	95.82 ± 1.27	95.10 ± 1.51
DDG-Mixstyle [36]	97.33 ± 0.21	97.02 ± 0.12
BoDA [24]	96.91 ± 1.03	96.27 ± 1.50
DM [32]	96.64 ± 0.74	95.83 ± 0.99
MTL [45]	97.22 ± 0.37	96.69 ± 0.37
IRM [40]	95.81 ± 1.31	95.03 ± 1.63
DaNet	97.40 ± 0.61	96.74 ± 0.74
DaNet*	**97.74 ± 0.41**	**97.21 ± 0.55**

The best results are **bold**.

Table 7
The average results of other scenarios.

Method	F1-micro (%)	F1-macro (%)
WBC-Benchmark [18]	91.68 ± 2.24	85.41 ± 3.34
DDG-Mixstyle [36]	92.55 ± 1.52	88.00 ± 2.21
BoDA [24]	95.03 ± 0.86	89.73 ± 1.48
DM [32]	93.83 ± 1.44	88.02 ± 1.85
MTL [45]	94.56 ± 1.76	89.45 ± 2.22
IRM [40]	85.36 ± 4.39	75.80 ± 4.93
DaNet	95.56 ± 1.00	90.51 ± 1.66
DaNet*	**95.67 ± 0.64**	**90.60 ± 1.08**

The best results are **bold**.

and F1-macro scores (Table 7). This demonstrates the strong generalization ability of our approach in WBC classification tasks.

2. In the scenarios where Acevedo-20 and Bodzas-23 were used as test sets, our method consistently achieved the highest scores, with F1-micro values of 93.47% and 97.74%, respectively. Although the highest score on the Matek-19 dataset was obtained by the DDG-Mixstyle method (with an F1-micro of 96.33%), our approach closely follows. These results effectively validate the adaptability and robustness of our method in WBC classification tasks.
3. Notably, the Matek-19 dataset exhibits obvious data imbalance problem. When this dataset was used as a training set, our method achieved the best results (Tables 5 and 6), indicating a potential advantage in handling imbalanced data.

4.5. Ablation study

Table 8 presents an ablation study comparing DDA and BMDM, allowing for an analysis of the contributions of each individual component to the overall performance. We provide a clear delineation of the usage status of DDA and BMDM, where the symbols ✓ and ✗ represent the inclusion and exclusion of these methods, respectively. Additionally, the first column is designed to clearly indicate the variants of the components, with the following interpretations for each variant:

Table 8
The results of the ablation study.

Variants	DDA	BMDM	F1-micro(%)	F1-macro(%)
M1	✗	✗	91.57±2.28	89.01±2.50
M2	✓	✗	92.37±0.97	90.29±0.92
M3	✗	✓	91.97±2.13	90.65±2.15
M4	✓	✓	**93.47±1.79**	**92.17±2.20**

The best results are **bold**.

- **Variant M1**: Using ResNet-50 as the backbone for Empirical Risk Minimization [55].
- **Variant M2**: Variant M1 with BoDA, setting $\omega = 0.1$ and $\nu = 0.5$.
- **Variant M3**: Variant M1 with BMDM, setting α to 0.1.
- **Variant M4**: Variant M3 with $\mathcal{L}_{DDA}$, setting $\omega = 0.1$ and $\nu = 0.5$.

Our ablation study uses the Matek-19 and Bodzas-23 datasets as training sets and the Acevedo-20 dataset as an independent test set to evaluate generalization performance. As shown clearly in Table 8:

1. The individual introduction of DDA and BMDM techniques can each enhance generalization capability to a certain extent. Specifically, DDA demonstrates relatively better performance in terms of the F1-micro score, with an increase of 0.8%, while BMDM shows stronger results on the F1-macro metric, with an improvement of 1.64%. This suggests that BMDM is able to better balance performance across different classes during the training process, preventing the model from being overly biased towards the majority classes and neglecting the minority ones.
2. The combination of DDA and BMDM leads to further improvements, with the F1-micro and F1-macro scores increasing by 1.1% and 1.88%, respectively, compared to the use of DDA alone. This indicates that BMDM, as a data augmentation technique, can further and more effectively enhance distribution alignment.

4.6. The weight of DaNet loss function

In Eq. (5), the value of ω was empirically set to 0.1, which resulted in relatively good average F1-micro scores. The DaNet method was used to train models with different ω hyperparameters. Each set of experiments was repeated five times, and the average F1-micro score on the test set was calculated for each group. Fig. 3 presents the experimental results for different ω values across three training scenarios. The different lines represent the impact of varying ω values on the performance of various unseen datasets, with the AVG line showing the average of the other three results. We observed that the value of ω has a noticeable impact across different scenarios. For instance, $\omega = 0.4$ performs relatively well when Matek-19 is used as the test set, whereas $\omega = 0.1$ yields better results on Acevedo-20. Although $\omega = 0.1$ is not optimal in every scenario, it provides a relatively balanced performance overall. Our hyperparameter selection strategy is designed to adapt to a wider range of scenarios rather than optimizing for a specific one.

4.7. Effectiveness of data distribution alignment

To evaluate the effectiveness of DaNet in distribution alignment compared to other methods, we employ t-SNE [56] visualization techniques to examine the distribution alignment in models trained with different approaches. The t-SNE technique enables nonlinear dimensionality reduction of the features extracted by the model, allowing for the visualization of these reduced data points and facilitating the identification of clusters within the data. Ideally, in a well-aligned data distribution, features from different classes should exhibit clear

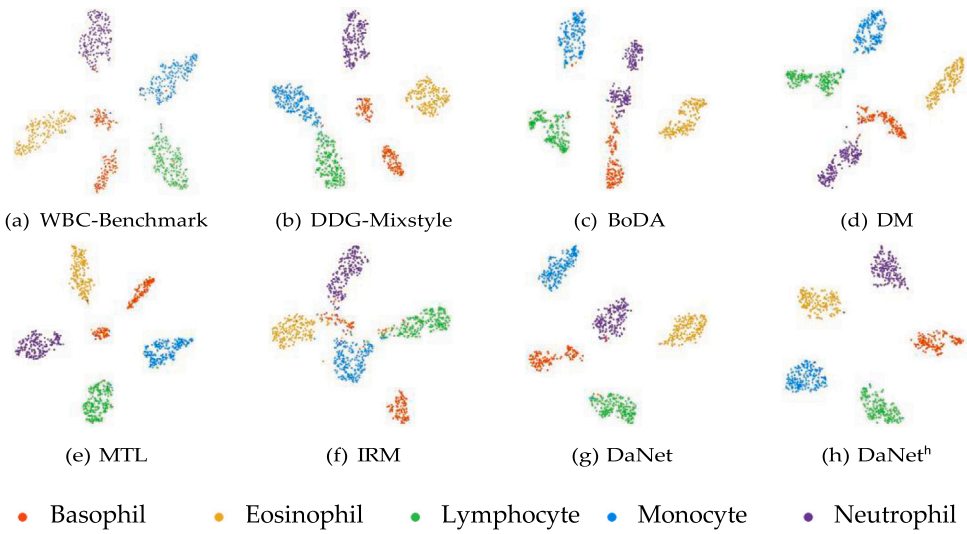

Fig. 2. Subfigures (a–g) are t-SNE feature distribution visualizations based on features extracted from different models. Each model was trained using Matek-19 and Bodzas-23 as the training sets, and up to 100 random samples from each source domain were selected for t-SNE dimensionality reduction and visualization. Subfigures (h) and (g) were obtained using the DaNet method, with features extracted from models trained in entirety mode and balance mode, respectively. Subfigures (a–f) are based on other baseline methods.

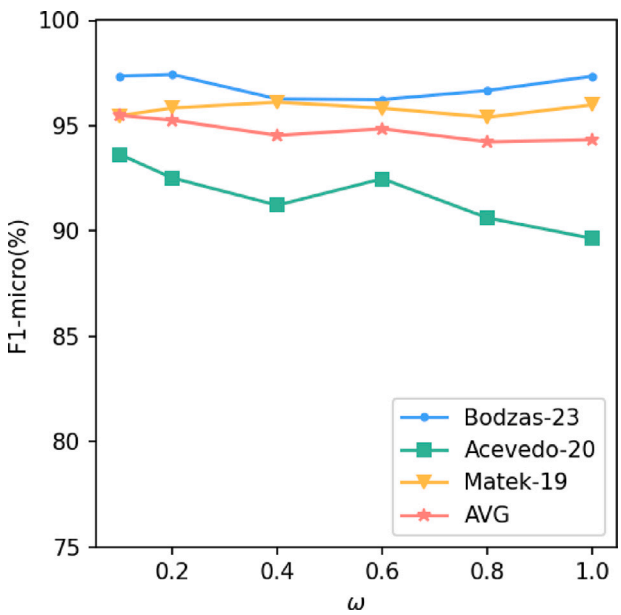

Fig. 3. The vertical axis represents the average F1-micro score of the models trained using the DaNet method in entirety mode, and the horizontal axis represents the weight (ω) of $\mathcal{L}_{DDA}$, consistent with the value in Eq. (5). Different lines represent the performance on different unseen datasets. The red AVG line is the average of the other three results.

decision boundaries, while features from the same class should be tightly clustered. However, the challenge in WBC classification arises from the subtle differences between classes and the significant data imbalance in certain datasets (e.g., Matek-19), which complicates the establishment of decision boundaries and the learning of discriminative features.

In Fig. 2, we use the Matek-19 and Bodzas-23 datasets as the training sets, randomly selecting up to 100 samples per class from each domain in the training set, and extract the features of these samples for t-SNE dimensionality reduction visualization. Each subfigure represents the t-SNE visualization of the training samples based on the models trained using different methods. We first examine the data distribution achieved by the BoDA technique (Subfigures c). Although the data distributions for each class are relatively concentrated, there are pronounced gaps between several classes. A similar issue is observed with the DM method, suggesting that common features across different classes were not effectively learned. In other baselines (e.g., MTL, Subfigures e), the Basophil class is split into two distinct clusters, primarily due to the disparity in the number of Basophil samples across the two source domains, which hampers effective feature alignment. In contrast, our methods (DaNet and DaNet*) result in more concentrated feature distributions for each class when compared to other methods, particularly in the case of DaNet*, which is more favorable for learning distinctive WBC features. We attribute the effectiveness of distribution alignment achieved by DaNet and DaNet* to the introduction of nonlinear data perturbations through BMDM during the training process. This perturbation enables the model to effectively avoid learning misleading shared features—features present in the current source domains for a particular class but not truly discriminative for that class.

4.8. Discussion

In prior research [32], it was demonstrated that employing a pixel-level mixup scheme encourages the domain discriminator to uncover the intrinsic structures of different domains, effectively reducing discrepancies between them. Building upon this concept, our proposed BMDM enhances the learning of distinctive features across various source domains. Additionally, the refined sampling strategy employed in BMDM adeptly addresses imbalanced data scenarios, improving the model's ability to generalize across diverse classes. This enhancement is illustrated in the comparison of subplots (g) and (h) in Fig. 2, where even classes with significant quantity disparities, such as basophils, manage to establish clear decision boundaries.

When BMDM is integrated with Domain Distribution Alignment (DDA), the resulting combination demonstrates superior performance relative to the latest domain generalization methods. This affirms the robustness and stability of our enhanced distribution alignment strategy in WBC classification tasks. Specifically, our DDA approach minimizes intra-class differences while amplifying inter-class distinctions across different source domains, effectively eliminating redundant and non-essential parameters. This allows the model to refine its parameters for improved generalization [57]. Furthermore, through ablation studies and t-SNE visualization experiments, we have shown that our method significantly enhances the learning of distinctive features compared to the BoDA algorithm. This enhancement is critical for maximizing the

effectiveness of distribution alignment, particularly in classifying WBC categories with subtle differences.

Experiment results show that DaNet achieved the best performance in scenarios where Acevedo-20 and Bodzas-23 served as test sets. However, when Matek-19 was used as the test set, our methods did not surpass the performance of DDG-Mixstyle. This discrepancy can be attributed to the inherent imbalance in the Matek-19 dataset, which may have adversely affected our methods' performance. Specifically, while the average accuracy per class for DDG-Mixstyle on the Matek-19 dataset is 83.24%, DaNet and DaNet* achieved accuracies of 84.12% and 84.48%, respectively. During training, DaNet places a relatively greater emphasis on class balance, which may yield better results for smaller classes but slightly lower overall performance on imbalanced datasets.

Moreover, DaNet is designed to tackle more challenging issues related to data imbalance. However, legal restrictions and privacy concerns limit the availability of comprehensive WBC datasets, with most publicly accessible ones lacking complete class representations. To ensure fair cross-validation of cross-domain generalization performance, we specifically selected common classes from each dataset. We focused our study on WBCs due to personnel and equipment constraints, but our approach is also applicable to other tasks facing similar challenges. While DaNet demonstrates promising performance improvements over other recent DG methods for WBC classification, there is still room for refinement in the learned features. Our future work will concentrate on better capturing discriminative features to further enhance the effectiveness of distribution alignment.

5. Conclusion

In this study, we present DaNet, a novel approach that synergistically combines data augmentation through BMDM and DDA. The primary objective of DaNet is to enhance data quality and achieve balanced sampling, which significantly boosts the effectiveness of DDA in extracting distinctive features from WBC images. To assess the performance of our method in the context of domain generalization, we conducted experiments using three publicly available WBC datasets: Matek-19, Acevedo-20, and Bodzas-23. From each dataset, we selected five common classes for comparison. Our methodology was benchmarked against several recently proposed DG techniques, highlighting its competitive edge. Through comprehensive ablation studies, including distribution alignment verification and baseline comparisons, we demonstrated that DaNet not only achieves superior distribution alignment but also enhances generalization performance in WBC classification tasks. Furthermore, we introduced DaNet*, an adaptation of DaNet that modifies the sampling strategy to establish clearer feature boundaries, further improving classification accuracy.

CRediT authorship contribution statement

Wenpeng Gao: Writing – original draft, Visualization, Methodology, Investigation, Data curation. **Liantao Lan:** Writing – review & editing, Validation, Supervision. **Xiaomao Fan:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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